Chapter 03 Vector and Two-Dimensional Motion

Properties of Vectors; Computation of Vectors

Omits. Learned in mathematics.

2D Displacement

The position of an object is described by its position vector, $\vec{r}$. The displacement is defined as the change in its position.

$$\Delta\vec{r} = \vec{r}_f - \vec{r}_i$$

Velocity

Average velocity is the ratio of the displacement to the time interval for the displacement.

$$\vec{v} = \frac{\Delta\vec{r}}{\Delta t}$$

The instantaneous velocity:

$$\vec{v} = \lim_{\Delta t \to 0} \frac{\Delta\vec{r}}{\Delta t} = \frac{d\vec{r}}{dt}$$

Acceleration

The rate where the velocity changes

$$\vec{a} = \frac{\Delta\vec{v}}{\Delta t}$$

The instantaneous velocity:

$$\vec{a} = \lim_{\Delta t \to 0} \frac{\Delta\vec{v}}{\Delta t} = \frac{d\vec{v}}{dt}$$

Acceleration can result from change in speed or change in direction.

Projectile Motion

Move in both two directions simultaneously, following a parabolic path.

Relative Velocity

Relate the measurements of two different observers.

A moves relate to B:

$$\vec{v}_{AB} = \vec{v}_{AE} - \vec{v}_{EB}$$